

Hazem Hassan

Senior AI/ML Engineer — Computer Vision · Production ML

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PROFESSIONAL SUMMARY

Senior Computer Vision Engineer with 9+ years taking vision systems from research to field-deployed product. Hands-on with thermal/IR image processing (defogging/dehazing, optical-flow anomalous-motion detection), multi-object tracking, and low-latency deployment on embedded NVIDIA hardware (Jetson, TensorRT, ONNX), alongside deep document-vision work — layout analysis at 98% F1, complex graphics-component analysis and understanding at 93% F1, and Arabic OCR systems shipped to production. First place in the ICFHR 2018 RASM international manuscript-recognition competition. Electrical & Electronics Engineering background, with a track record of optimizing pipelines for resource-constrained edge devices and leading cross-functional hardware-software integration. Comfortable benchmarking and hardening trackers for dynamic, real-world conditions.

ROLES AND SPECIALTIES

Roles: Senior ML Engineer, AI Technical Lead, Senior Computer Vision Engineer.

Specialties: Computer Vision Pipelines, OCR, Video Analytics, Multi-Object Tracking, Action Recognition, Automated Labeling, Human-in-the-Loop Workflows, Active Learning, Weak Supervision, Dataset Quality Assurance, ML APIs, MLOps, Cloud/Edge Deployment.

CORE SKILLS

Programming: Python (expert); C/C++ (intermediate); CUDA / NVIDIA platforms

Thermal / IR & Image Processing: Thermal/IR image processing, defogging & dehazing, image enhancement & restoration, denoising, optical flow, motion & anomaly detection (familiar with NUC / bad-pixel-correction concepts – ready to ramp on sensor-level calibration)

Object Tracking: Multi-object tracking (MOT), feature-based & optical-flow tracking, deep-learning trackers, ID-switch reduction, tracker tuning for dynamic scenes

Real-Time & Embedded: NVIDIA Jetson, real-time multi-threaded pipelines, low-latency inference, model optimization for constrained hardware, TensorRT, ONNX

ML / Deep Learning: PyTorch, TensorFlow, model training & fine-tuning, custom loss design, evaluation metrics, error & failure analysis

Data Engineering: SQL, PostgreSQL, NoSQL Basics, Pandas, NumPy, Data Cleaning/Transformation, Large-Scale Image/Video/Text Processing, Dataset Preparation Pipelines, Feature Engineering.

Deployment & MLOps: Model Serving, FastAPI, REST APIs, Docker, Kubernetes, CI/CD, Model Optimization, ONNX, TensorRT, Inference Performance Tuning, Monitoring & Failure Analysis.

Cloud & Edge: GCP (Vertex AI), AWS (SageMaker / Ground Truth), NVIDIA Jetson, Real-time Inference, Low-latency Video Pipelines.

Leadership: Technical Leadership, System Design, Cross-functional Collaboration, Mentoring, Code Reviews.

WORK EXPERIENCE

Senior ML Engineer

Beyond AI

March 2024 — Present, Remote, EMEA

Computer Vision

- Built an in-house visual-prompt detector (region-prompt contrastive alignment) that adapts to new domains from a few reference images, with no re-training or re-labeling — +3 AP over YOLOE.
- Fine-tuned OpenVLA for task-specific robotic manipulation and quantized the model for edge inference, cutting memory footprint and latency enough to run the policy on-device rather than off-loading to a server.
- Developed thermal/IR image-processing algorithms: fog dehazing/defogging to restore visibility in low-visibility conditions, and optical-flow-based detection of anomalous motion on IR camera feeds.
- Deployed real-time video analytics on NVIDIA Jetson edge devices across 10+ camera feeds / 5+ sites, sustaining 25–35 FPS at <150 ms latency for tracking, counting, and behavior analytics.
- Tuned detection + multi-object tracking for “Easier Gates” flow analytics, reaching 94% flow accuracy and reducing ID switches by 14%.
- Optimized vision pipelines for resource-constrained edge hardware using TensorRT/ONNX, balancing accuracy against latency and compute budgets.
- Built PPE-compliance detection (Aramco) at 94% precision / 90% recall, cutting false alarms 25% via post-processing and threshold calibration.
- Designed active-learning data-selection loops prioritizing high-uncertainty and edge-case samples, cutting annotation effort while improving robustness in dynamic scenes.

Document AI & OCR

- Built layout analysis and complex graphics-component analysis for Arabic documents — 98% F1 on layout regions, 93% F1 on graphics components, and 84% on graphics reasoning/understanding (chart and diagram semantics, figure-caption linkage).
- Fine-tuned PaddleOCR for Arabic text recognition — 14% WER general and ~8% WER on domain-specific documents— handling cursive script, ligatures, and right-to-left layout

Technical Lead

IDefy (Digital Identity/eKYC)

October 2022 — February 2024, Giza, Egypt

- Led digital identity onboarding products (SaaS/SDK/web/mobile), delivering 3 production deployments end-to-end.
- Built validation and quality-control workflows for OCR, face matching, liveness, and identity-verification datasets, improving verification success rate by 20% through data quality tuning and systematic failure analysis.
- Developed configurable OCR/document-processing pipelines that adapted to new structured documents using limited samples, annotation rules, and validation checks, reaching 96% recognition accuracy.
- Analyzed production failure cases across OCR, face verification, and liveness detection to identify noisy data, poor image quality, edge cases, and model retraining opportunities.

Senior ML Engineer (Computer Vision)

RDI

January 2021 — October 2022, Giza, Egypt

- Led the delivery of the Sotoor Arabic OCR System, achieving 12% WER on Arabic manuscripts.
- Built document OCR pipeline modules (layout analysis, segmentation, font detection, recognition, denoising, correction), reducing OCR error by ~7% on noisy scans.
- Adapted the K2 Toolkit for OCR training/decoding, improving training efficiency by ~25% and gaining ~4% WER, boosting sequence recognition stability.

ML Researcher

RDI

January 2017 — January 2021, Giza, Egypt

- Collaborated on a team building NLP systems for the Arabic language across speech and text.
- Developed and deployed production systems, including Speech Recognition, Optical Character Recognition, Text Diacritization, Voice Conversion, and Quran Recitation Verification.

PROJECTS

Agentic Agent Call Center

- Built an agentic AI call-center assistant using LLM tool-calling and Multi-RAG retrieval to automate intent handling, answer generation, and CRM/ticketing actions.
- Designed a reliable conversation orchestration layer with guardrails, escalation-to-human routing, and quality evaluation to ensure safe and consistent customer support automation.

Fashimi APP — Lead Engineer

- Built an AI try-on platform using LLMs for recommendation + prompt generation and diffusion models for avatar generation and photorealistic try-on rendering.
- Developed a recommendation engine leveraging mood/occasion/climate/user preferences to produce personalized outfit prompts.

Valorant Overlay — Developer

- Designed a real-time multi-threaded system extracting gameplay data from Valorant for esports competition overlays.
- Implemented object detection, pattern recognition, and color analysis, achieving a response latency of 50-72 msec.

EDUCATION

Bachelor of Science in Engineering

Cairo University — Electrical Electronics and Communication Engineering • 2016

PUBLICATIONS

Arabic Multi-Genre Broadcast Speech Recognition — Conference Paper, December 2019 — IEEE ASRU 2019
Prognostic AI for Post-Operative Infections — Conference Paper, November 2023 — FCI Annual Conference
QuranScript Verification System — Conference Paper, November 2016 — Sixteenth ESOLE Conference of Language Engineering

HOBBIES

Swimming

Gaming

Padel